

NILAY VATSAL

Hyderabad, Telangana

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EDUCATION

International Institute of Information Technology, Hyderabad

August 2024 – June 2026

M.Tech. in Computer Science and Information Security - CGPA - 8.33

Hyderabad, India

Bharati Vidyapeeth(DU) Department of Engineering and Technology

Nov 2020 - June 2024

B.Tech in Computer Science and Engineering - CGPA - 9.2

Mumbai, Maharashtra

Experience

Finzie by Samwaad | Python, NLTK

July 2023 - August 2023

Software Developer Intern

Remote

- Developed an SMS parser using NLTK to automatically extract and structure financial transaction data.
- Analyzed spending patterns from over 10,000 users to build the logic for a personalized budget alert system.
- Created an engine to accurately classify user expenses into various spending categories.

Projects

Version Control System | C++

- Built a core VCS by implementing the **Git object model** (blob, tree, commit) to track file history.
- Applied **SHA-1 hashing** for content addressing and data **integrity**, and implemented **file compression** to optimize storage.
- Developed a **command-line interface (CLI)** supporting key operations like **init**, **add**, **commit**, and **checkout**.

Distributed File System | C++ with MPI

- Engineered a **distributed file system** in C++ with MPI, implementing **chunk replication** and **fault tolerance** for **high availability**.
- Implemented a **dynamic load balancing** algorithm to optimize **storage efficiency** and **data access speed** across nodes.
- Designed robust **failover and recovery protocols** ensuring **data consistency** during node failures.
- Utilized **mutex locks** for **atomic operations**, ensuring **concurrency control** and **data consistency**.

SecureConnect | Python

- Constructed a **secure, concurrent client-server system** using **Diffie-Hellman** and **DDES** for **end-to-end encryption**.
- Integrated **session tokens** and **HMAC validation** to prevent **replay attacks** and ensure **data integrity**.
- Implemented robust **error handling** for cryptographic operations, including key exchange, token validation.

TECHNICAL SKILLS

C/C++, Python, gRPC, SQL, Git/Github, Linux, Flask, OOPS, OS, DBMS

ACHIEVEMENTS

GATE Exam

- GATE 2024 - 98.5 percentile
- GATE 2023 - 96.5 percentile

Competitive Programming

- Specialist at Codeforces (Max rating: 1423)
- 3 Star at Codechef (max rating: 1686)